Drive for better vision

EVK and PC Tool User Guide for HX6538 AIoT EVB V1.2

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Agenda

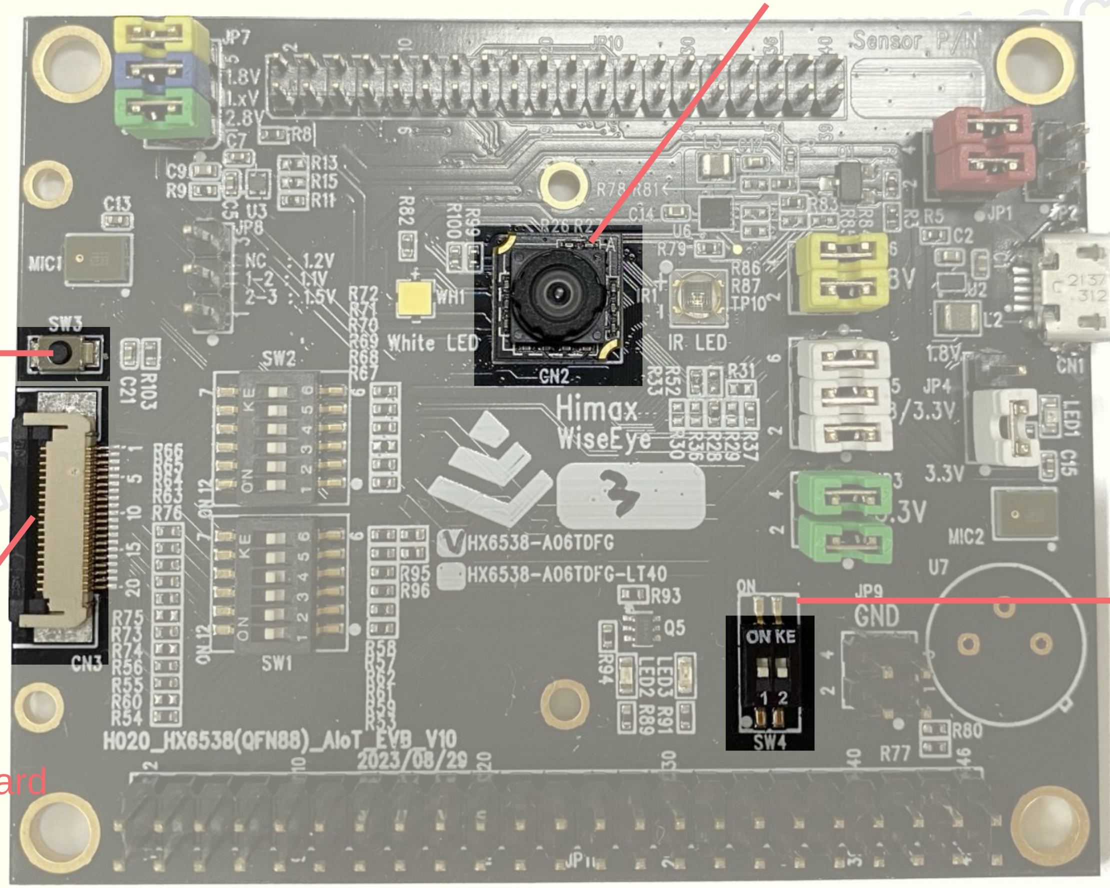
- **Board Introduction**
  - EVB Introduction
  - Debug Board Introduction
- **PC Tool Usage**
  - PC Environment Setting
  - Flash Programming Procedures
  - Image Reception and Result Detection

Board Introduction

HX6538 AIoT EVB

CIS sensor

Reset button

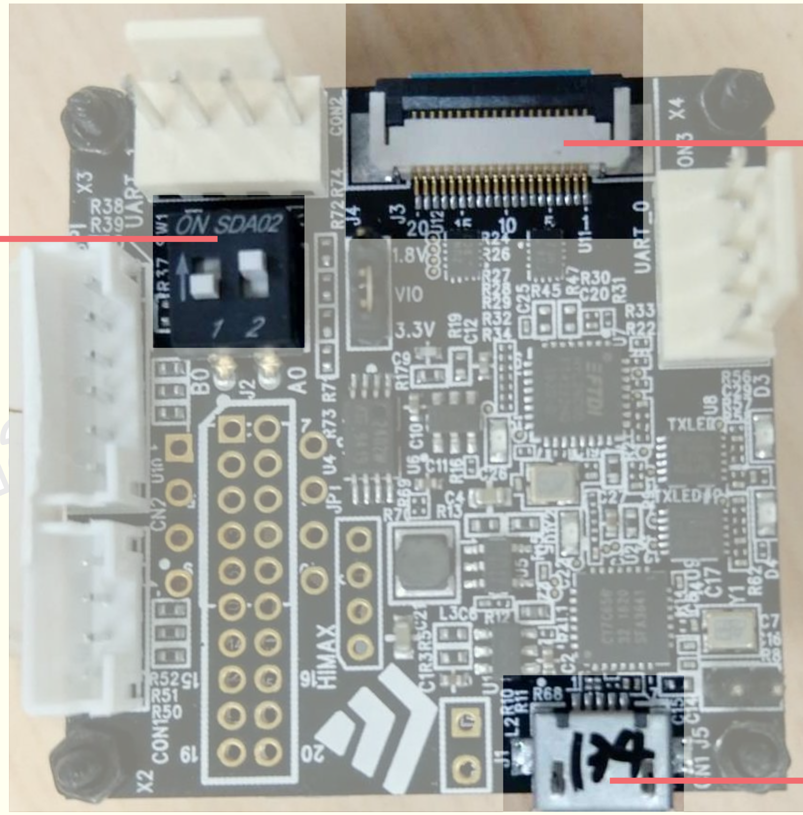


Boot option

Connecter:
Connect to Debug Board

Debug Board (FT4222H)

SPI Mux Pin select

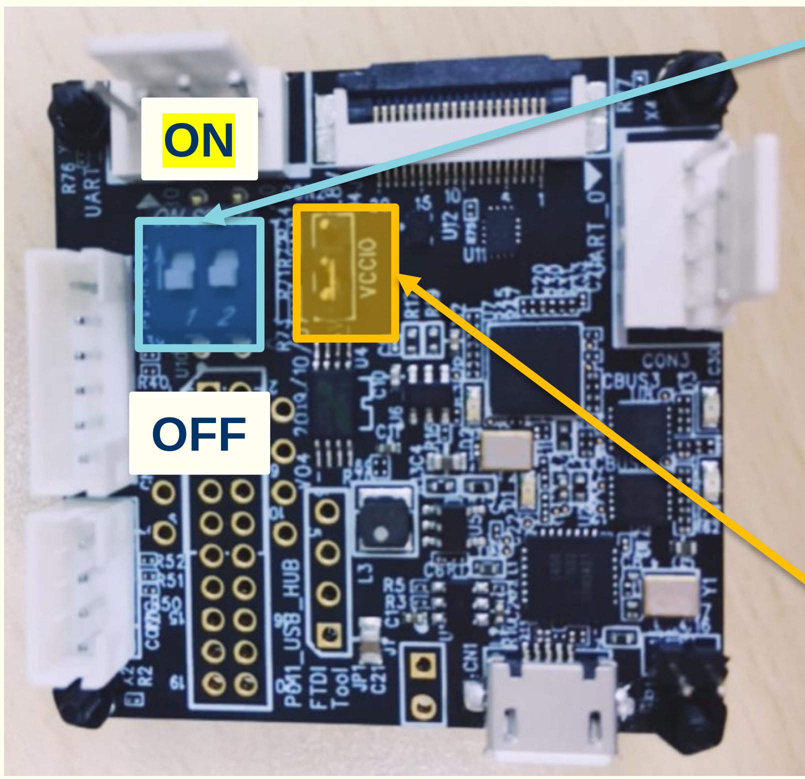


Connecter:
Connect to EVB

USB connecter:
Connect to PC USB port

Debug Board (FT4222H)

- **SPI Mux Pin select**
  - **[1, 2] = [OFF, ON]:** FT4222H SPI Slave mode for image transfer
  - **[1, 2] = [ON, OFF]:** FT4222H SPI Master mode for image transfer



SPI Pin option select

SWITCH	MODE
ON OFF 1 2 [OFF, ON]	**FT4222H:** SPI Slave **EVB:** TRANSMIT_PROTOCOL = SPI_MASTER
ON OFF 1 2 [ON, OFF]	**FT4222H:** SPI Master **EVB:** TRANSMIT_PROTOCOL = SPI_SLAVE

Voltage Switch
1.8V
3.3V

PC Tool Usage

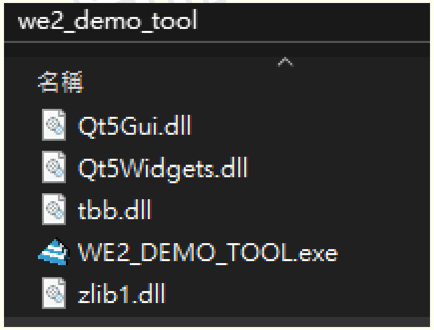
(*WE2_DEMO_TOOL*)

PC Tool Usage *(WE2_DEMO_TOOL)*

- **PC Environment Setting**
  1. **Install FT4222H driver**
     (path: SDK_root/_Documents/driver.7z)

  2. **Install VC++ Distribution**
     (Download VC_redist.x64.exe by url:
     https://learn.microsoft.com/en-US/cpp/windows/latest-supported-vc-redist?view=msvc-170)

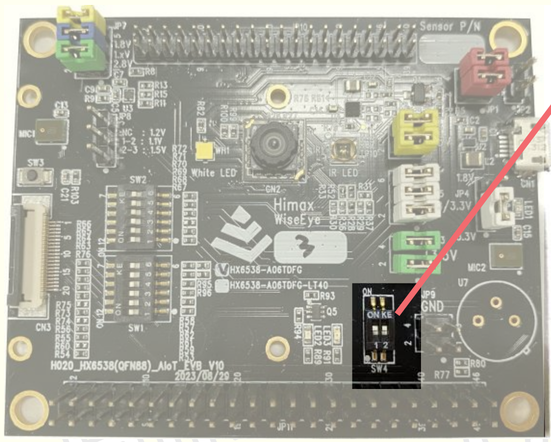
- **Execute *WE2_DEMO_TOOL.exe***
  (path: SDK_root/we2_demo_tool/WE2_DEMO_TOOL.exe)

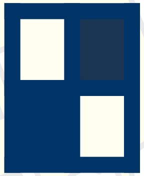


Flash Programming Procedures

EVB Flash Programming Setting

- **Set the switches on EVB board for flash mode.**
  - EVB switch **[1, 2] = [ON, OFF]**



SWITCH SETTING	MODE
 ON OFF 1 2 [ON, OFF]	**EVB:** Flash mode

EVB Flash Programming Procedures

1. **Launch the PC Tool “*WE2_DEMO_TOOL*”**
   (path: SDK_root/we2_demo_tool/WE2_DEMO_TOOL.exe)

2. **Following the procedures to flash firmware:**
   ① Switch to ***“BURN FLASH”*** page.

   ② Click the ***“Select File”*** to select flash image file.
   (path: SDK_root/we2_ota_tool/img/output.img)

   ③ Change Flash Type ***“WINBOND”***.
   ④ Click the ***“Start”*** to start burning image.
   ⑤ Wait the progress until 100%.

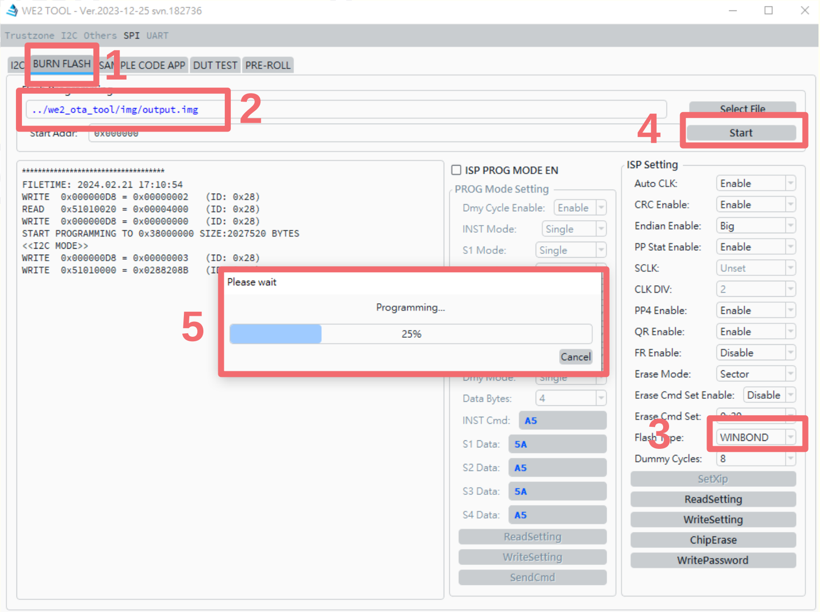
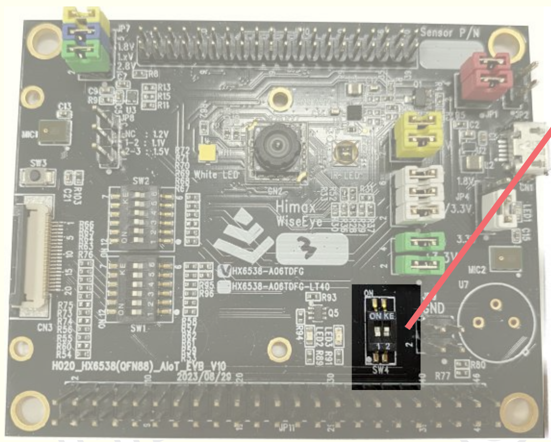


Image Reception and Result Detection

EVB Normal Mode Setting

- **Set the switches on EVB board for normal mode.**
  - EVB switch **[1, 2] = [OFF, OFF]**



SWITCH SETTING	MODE
ON OFF **1 2** **[OFF, OFF]**	**EVB:** Normal mode

Image Reception and Result Detection

- **Receive Image + Metadata**
  - ① Switch to **“*SAMPLE CODE APP*”** page.
  - ② Choose ***“SPI SLAVE”.***
  - ③ Enable **“*Enable Metadata*”.**
  - ④ Click ***“START”*** to start receiving the frame and meta data from EVK.
  - ⑤ PC Tool will show the image.
    (If any person was detected, PC tool will mark the detected person with bounding box.)